Faculty of Digital Engineering

Department of Computer Science

Specialized in Software Engineering

Capstone Proposal

Student Name	*Von visal*
Group	*Group 07*
Project Title	*Khmer-English Code -Switching Speech Dataset*
Started Date	*1st October 2026*
Ended Date	*30th March 2027*

***Abstract*—** This project builds an open Khmer-English code-switching speech dataset: recorded, transcribed, and language-tagged speech in which speakers mix Khmer and English within the same sentence. It targets atleast 60 validated hours by March 2027, released with a datasheet and baseline speech-recognition results. The dataset gives researchers a benchmark for a speech pattern that Khmer speech systems are rarely tested on.

***Keywords*—** code-switching, Khmer, English, speech corpus, automatic speech recognition, low-resource languages

I. Project Description

Khmer speakers often mix English words into Khmer sentences, yet speech data that captures this habit is scarce, so recognition systems are rarely trained or tested on it.

Problem statement. Cambodians commonly switch between Khmer and English within a single sentence, especially when talking about technology, business, and education. Khmer is a low-resource language for speech technology. Code-switching corpora exist mainly for other language pairs, such as Mandarin-English [2], and general resources such as Common Voice [3] and the Whisper training data [4] are not designed around mixed Khmer-English speech. We have not found a public Khmer-English code-switching corpus; the literature review in October 2026 will confirm this.

Project goal. Build, validate, and publicly release a Khmer-English code-switching speech dataset with transcripts, language tags, and baseline recognition results.

Scope. The project covers recording, transcription, tagging, quality control, and baseline evaluation. It does not include building a production speech-recognition system.

Benefits.

- Researchers get a benchmark for code-switching recognition in a low-resource language.
- Cambodian developers can build voice tools, such as captioning and transcription, that match how people actually speak.
- The team gains practice in data collection, annotation, and evaluation, and documents the data with a datasheet [1] so others can reuse it.

II. Project Objective

The project has five SMART objectives, each with a measurable target and a deadline (Table I).

TABLE I. SMART OBJECTIVES

No.	Objective	Measure of success	Deadline
O1	Define scope and annotation rules	Review of at least 10 corpora or papers; annotation guideline v1.0 approved by the advisor	30 Nov 2026
O2	Collect speech from diverse speakers	At least 30 consenting speakers and 80 hours of raw audio; neither gender above 60% of speakers	31 Jan 2027
O3	Transcribe and language-tag the speech	At least 60 hours transcribed and tagged; Cohen's kappa of 0.80 or higher on a 10% double-annotated sample	28 Feb 2027
O4	Evaluate the dataset with baseline models	Character error rate (CER) and word error rate (WER) of at least one pretrained model reported on the test split	15 Mar 2027
O5	Document and release the dataset	Datasheet, README, and license published with the data; final report submitted	30 Mar 2027

Targets are proposed values, to be confirmed with the advisor.

III. Methodology/Activities

The work runs in eight steps, from literature review to public release, and each step ends with a deliverable.

1. **Review and scope (Oct 2026).** Survey existing code-switching corpora and Khmer speech resources [2], [3], and define the types of code-switching to capture. Deliverable: review notes and scope statement.
2. **Guidelines and tools (Oct–Nov 2026).** Write the annotation guideline, a consent form, and recording prompts; set up the recording tool and an annotation tool such as ELAN or Label Studio. Deliverable: guideline v1.0.
3. **Pilot recording (Nov 2026).** Record 5 speakers, annotate their audio, then revise the guideline and prompts. Deliverable: pilot report.
4. **Main recording (Dec 2026–Jan 2027).** Record at least 30 consenting speakers, mixing scripted prompts with spontaneous conversation on topics such as technology, study, work, and daily life. Deliverable: raw audio and speaker metadata.
5. **Transcription and tagging (Dec 2026–Feb 2027).** Transcribe in Khmer script with English words in Latin script, and tag the language of each word. Double-annotate 10% of the audio to measure agreement with Cohen's kappa. Deliverable: validated transcripts.
6. **Quality control and splits (Feb–Mar 2027).** Remove clipped, noisy, or duplicated recordings, then split into train, dev, and test sets by speaker so no speaker appears in two sets. Deliverable: cleaned dataset with fixed splits.
7. **Baseline evaluation (Mar 2027).** Evaluate pretrained Whisper [4] and fine-tune a pretrained wav2vec 2.0 model [5] on the training split. Report CER as the primary metric, since Khmer text has no spaces between words, and WER with a word segmenter, separately for code-switched and monolingual utterances. Deliverable: baseline results table.
8. **Documentation and release (Mar 2027).** Write the datasheet [1], README, and license terms, publish the dataset, and submit the final report. Deliverable: public release and final report.

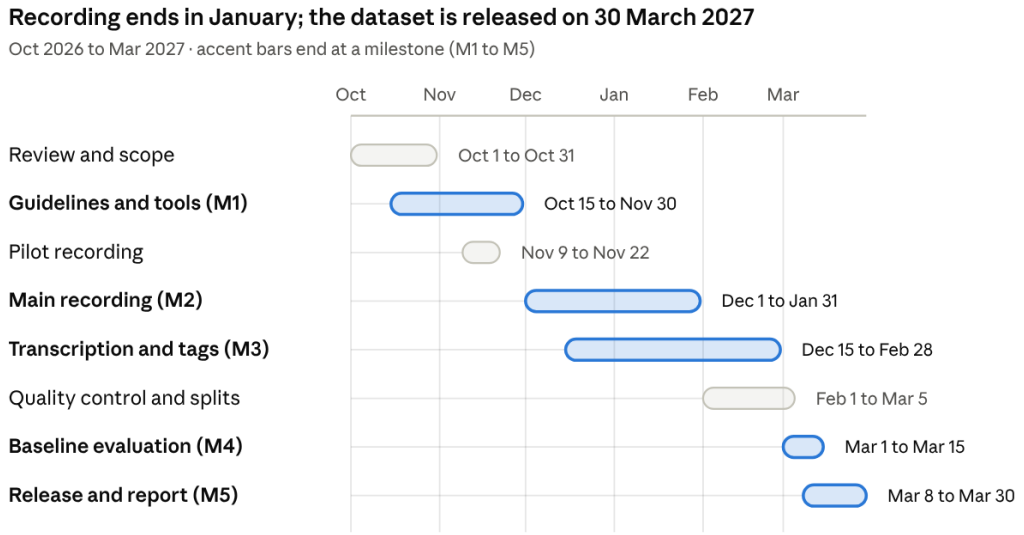
TABLE II. PLANNED DATASET SPECIFICATION

Property	Planned value
Speakers	At least 30 adults, mixed ages and regions where possible
Speech volume	At least 80 hours recorded; at least 60 hours validated
Speech types	Scripted prompts and spontaneous conversation
Audio format	16 kHz, 16-bit, mono WAV
Transcripts	Khmer script, English words in Latin script, language tag per word
Splits	Train, dev, and test by speaker, roughly 80% / 10% / 10%
Metadata	Speaker ID, gender, age group, region, device, recording environment

Property	Planned value
Privacy	Informed consent; speakers identified by ID only, no names stored
License	Open license, chosen with the advisor

IV. Project Milestone

Five milestones (M1–M5) fall between October 2026 and March 2027, ending with the public release on 30 March 2027 (Fig. 1).



project schedule · 8 activities, 5 milestones

Fig. 1. Project schedule, October 2026 to March 2027. M1: annotation guideline v1.0 (30 Nov 2026). M2: recording complete (31 Jan 2027). M3: annotation complete (28 Feb 2027). M4: baseline results (15 Mar 2027). M5: release and final report (30 Mar 2027).

References

1. T. Gebru et al., "Datasheets for datasets," *Communications of the ACM*, vol. 64, no. 12, 2021.
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5. A. Baevski, H. Zhou, A. Mohamed, and M. Auli, "wav2vec 2.0: A framework for self-supervised learning of speech representations," in *Proc. NeurIPS*, 2020.

Seen and approved by advisor: _____